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Deep Learning based Modeling and Inverse Design for Arbitrary Planar Antenna Structures at RF and Millimeter-Wave

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***Abstract*—In this paper, we introduce inverse design of nearly arbitrary planar antenna structures with a deep convolutional neural network (CNN) modeling that allows rapid and accurate prediction of antenna performance (scattering parameters and radiation patterns). Quite distinct from prior efforts of ML-based antennas with fixed template geometries and finite degrees of freedom, this approach of generalizing to arbitrary planar structures opens up a new design space for antenna structures with properties beyond what can be achieved with antennas optimized from a finite library. By eliminating complex time consuming electromagnetic simulations with an ML-based approach, we propose an inverse design with evolutionary algorithms that allows a much larger search space than classical genetic algorithm based approaches. We demonstrate this methodology with simulation and measurement results of inverse designed compact, broadband and multi-band planar antennas operating at RF (2-5 GHz) and mmWave (20-40 GHz).**

***Index Terms*—Deep neural networks, antenna design, electromagnetic simulations, genetic algorithm, machine learning**

I. INTRODUCTION

Expansion of wireless technologies to new frequency bands and functionalities brings new design challenges and requirements for antenna engineers in terms of designing antennas across spectrum that are multi-band, compact, efficient, and has designer radiation properties [1]. Given the high non-convexity of the design space, prior approaches with evolutionary algorithms (such as genetic algorithm, particle swam optimization) have been limited in their search space and degrees of freedom due to the time consuming electromagnetic simulations. To relax this trade-off, data driven surrogate modelling of antennas has been proposed in the prior art [2], [3]. In majority of these work, starting point was parameterization of a known antenna structure. However, there is no reason to believe that limiting the design space to a template will yield the best possible performance. In this paper, we bring a hybrid inverse design approach by combining deep convolutional neural network (CNN) modelling of near-arbitrary geometries with an optimization algorithm for antenna synthesis. Hereby, input geometries are discretization of a metal surface over $N_X \times N_Y$ pixels which can be "on" (indicating metal) or "off" (indicating no metal). Moreover, feed location can be varied and placed on any of the pixels. When combined with an evolutionary algorithm (such as genetic algorithm in this case), the approach allows us to scan an exponentially large design space due to the rapid CNN-based emulator to synthesize antennas with designer characteristics (scattering properties and radiation patterns as shown in Fig. 1).

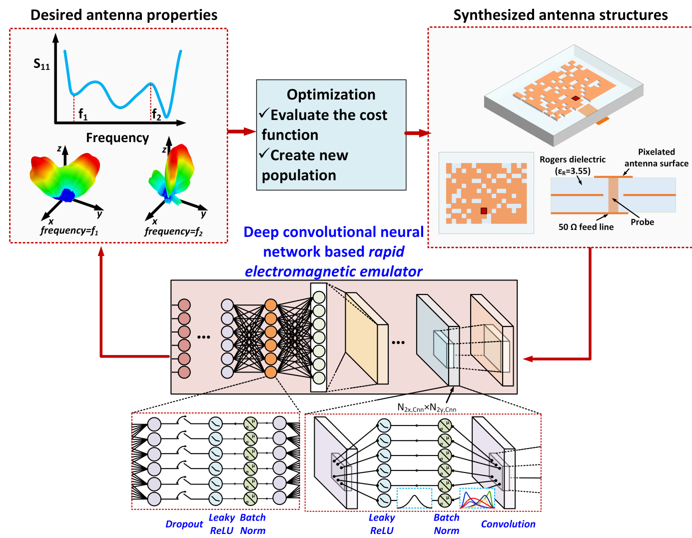

Fig. 1. Deep CNN-based forward modeling of antennas with arbitrary planar structures. Using this rapid and accurate emulator, the inverse synthesis opens up a large search space of antennas with designer radiation properties.

II. DEEP CNN-BASED MODELING AND INVERSE DESIGN

The overall inverse synthesis methodology for the desired radiation properties is shown in Fig. 1. We rely on a rapid and accurate forward model based on deep CNN, where the input is the geometry of the antenna and the output is the predicted performance (scattering parameters, radiation patterns). To model nearly arbitrary planar structures, we first discretize the metal region into pixelated structures that can be represented by a matrix. (Fig. 1). Given that the structure (and the image) itself is related to its radiating properties, we choose CNN as the natural machine learning model for this image-like feature set. Moreover, since CNN allows us to extract local features and relate to the radiation properties, and the solutions to Maxwell's equations are perturbed more strongly with local changes in boundary conditions, CNN is a suitable choice for this modeling. A CNN with different size of convolutional filters can capture this level of spatial dependence.

In this paper, we present this approach across RF (2-5 GHz) and mmWave (20-40 GHz). The top metal surface was divided into 12×12 pixels for both, allowing theoretically a design set of nearly $\sim 2^{144}$ ($\sim 10^{43}$) possible antenna structures. In addition to the planar metal surface, the feed position is a design variable and the model captures its effect as well. The metal layers and the dielectric are determined by the fabrication technology. For the training set, we use a training set of randomized structures (with augmentation) of a 0.7 million for RF and 1.1 million structures for the mmWave design. Datasets were roughly divided by 10:1:1 to training, validation, and test sets. While selecting the model, we performed training for different number of convolutional and fully connected layers to find out simplest model that can provide good test set prediction accuracy. As a result of this tuning, we arrived to an architecture with 9 convolutional layers each with 96 filters and 5 fully connected layers. Filter size for convolutional layers were gradually reduced from 8×8 to 3×3, in accordance to existing guidelines. Batch normalization layer and leaky ReLU activation function have been used after each convolutional and fully connected layers. In addition, a dropout layer follows the fully connected layers to avoid over fitting. Once converged, we deployed the network for antenna synthesis with different properties across RF and mmWave.

III. Results

To demonstrate the effectiveness of the modeling approach and the inverse design methodology, we present fabricated antennas and their measured properties in Fig. 2 and Fig. 3 in the RF and mmWave spectrum. Fig. 1 shows an example of a broadband patch antenna on a 12 mil RO4003C substrate operating between 30-33 GHz where the synthesized surface is an optimized pixelated antenna with an off-axis feed, as a result of the optimization algorithm supported by the CNN model (Fig. 1). The tight correspondence between the CNN modeling, the electromagnetic simulation and the measurement results shows the effectiveness of the modeling approach. As a point in comparison for this antenna with 3 GHz bandwidth (where S_{11} is below -10 dB), a traditional probe fed square patch antenna of the same size achieves 1 GHz impedance matching bandwidth around 32 GHz. Fig. 3 shows an example of a dual band RF patch antenna operating between 2-5 GHz and realized on a 60 mil RO4003C substrate. Even though a typical rectangular patch of the same sizes resonates around 4.1 GHz, the inverse antenna design allows us to create dual band properties where the antenna operating at 3.6 GHz is nearly 30% more compact.

IV. Conclusion

The presented approach of accurate and rapid modeling of arbitrary planar antennas with deep CNN and the inverse design can lead to a a new design space allowing designer radiation properties beyond the scope of antennas with optimized template-based geometries.

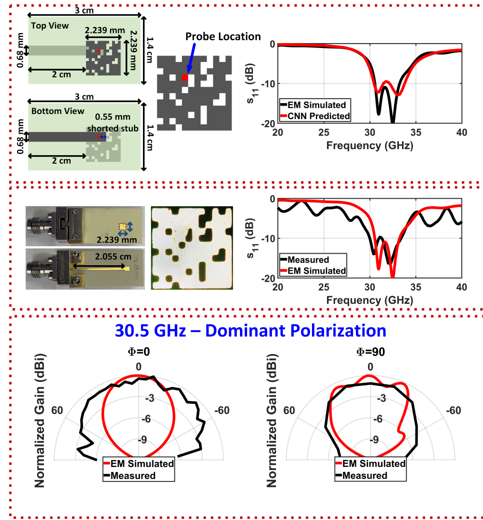

Fig. 2. Broadband mmWave microstrip antenna with an off-axis feed realized on a 12 mil RO4003C substrate. The design is synthesized with inverse method supported by the deep CNN modeling.

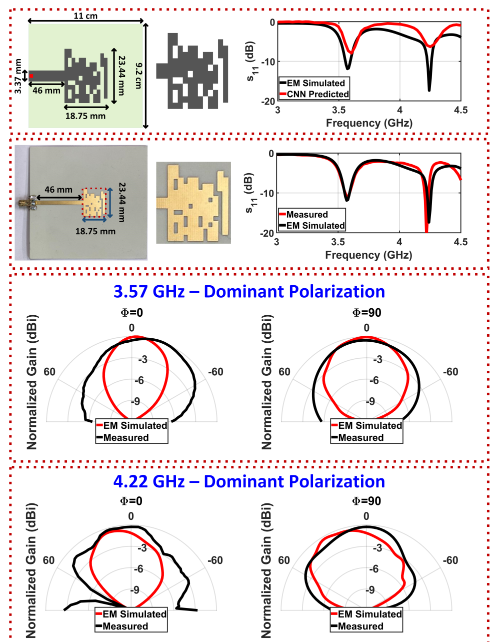

Fig. 3. Dual band compact RF antenna realized on a 60 mil RO4003C substrate. The design is synthesized with inverse method supported by the deep CNN modeling.

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